CS2211 Group Project

Design Documentation

Version	Date	Author(s)	Summary of Changes
1.0	2025.10.22	Peiyong Wang, Zhixian Wang, Xinyan Cai, Junqi Zheng, Jialin Li	Create introduction, class diagram, user interface mockup, and summary sections.

Table of Content

Section	Discription
1. Introduction	Overview of the problem being solved, the project's objectives, and references to important documents.
2. Class Diagrams	UML class diagrams illustrating main classes, attributes, methods, associations, and design reasoning.
3. User Interface Mockup	Static images and written explanation of the interactive Figma mockup, covering UI layout and design choices.
4. Summary	Brief wrap-up of the documentation, including definitions for terms, notations, or abbreviations used

Comments

1 Introduction

1.1 Overview

This project delivers a data-driven Java engine for classic point-and-click adventure games. At runtime, the engine loads a JSON or XML data file that fully defines the world's locations (name, description, picture, connections, objects), objects (name, description, attributes, optional contained items), and characters (phrases, desired objects/attributes), along with rules that govern interactions (use, give). Gameplay advances in discrete turns with a visible counter and optional turn limit. A graphical UI presents the current location, pictures, description, objects, connections, the player's inventory, command buttons, command output/feedback, starting message, turn count, and an exit path. Because content and logic are externalized, the same engine can run multiple adventures by swapping the data file.

Functionally, the system must support core commands—Go, Pick Up, Drop, Inventory, Examine, Use, Talk, Give—and apply rules to create and transform game state in response to player actions. Non-functionally, it must be implemented in Java (desktop), use a GUI toolkit (Swing or JavaFX), parse JSON/XML, provide Javadoc comments, give clear error feedback, perform responsively, and follow consistent team coding conventions.

1.2 Objectives

The project aims to develop skills in the following areas:

1. Applying the principles of software engineering toward a real-world problem.
2. Working with, interpreting, and following a detailed specification.
3. Creating models of requirements and design from such a specification.
4. Implementing the design in Java and dealing with decisions made earlier in the design process.
5. Creating graphical, user-facing content and applications.
6. Writing robust and efficient code.
7. Writing good, clean, well-documented Java code that adheres to best practices.
8. Reflecting on good and bad design decisions made over the course of the project.

1.3 References

1. CS2212A Group Project Specification - https://westernu.brightspace.com/content/enforced/127086-UGRD_1259_3953/CS2212A%20Project%20Specification.html
2. CS2211 Group Project Requirements Documentation (Group 2) -Gitlab, https://gitlab.sci.uwo.ca/courses/2025/09/COMPSCI2212/group02/-/wikis/Home/Requirements-Documentation
3. Maniac Mansion - Wikipedia, https://en.wikipedia.org/wiki/Maniac_Mansion
4. SCUMM - Wikipedia, https://en.wikipedia.org/wiki/SCUMM
5. Class diagram - Wikipedia, https://en.wikipedia.org/wiki/Class_diagram

Comments

2 Class Diagrams

Class Diagram Description

This UML class diagram depicts the core design of a point-and-click adventure game engine, emphasizing modularity and clear object-oriented layers. It cleanly separates game world data, player interaction, command processing, rule evaluation, and user interface into five main subsystems.

System Architecture

1. **Game Engine Layer** Controls game flow by processing player commands, applying rules, and managing turn progression.
2. **Game Data Layer** Defines the core data structures representing locations, inventory, entities and rules.
3. **Object and Rule Layer** Represents interactive game objects and encapsulates game logic through rules.
4. **Command Layer** Abstracts all player actions as commands, supporting validation and execution via the Command Pattern.
5. **User Interface Layer** Manages all visual output and player feedback.

Each subsystem interacts via well-defined interfaces to ensure loose coupling and high cohesion.

Key Classes and Responsibilities

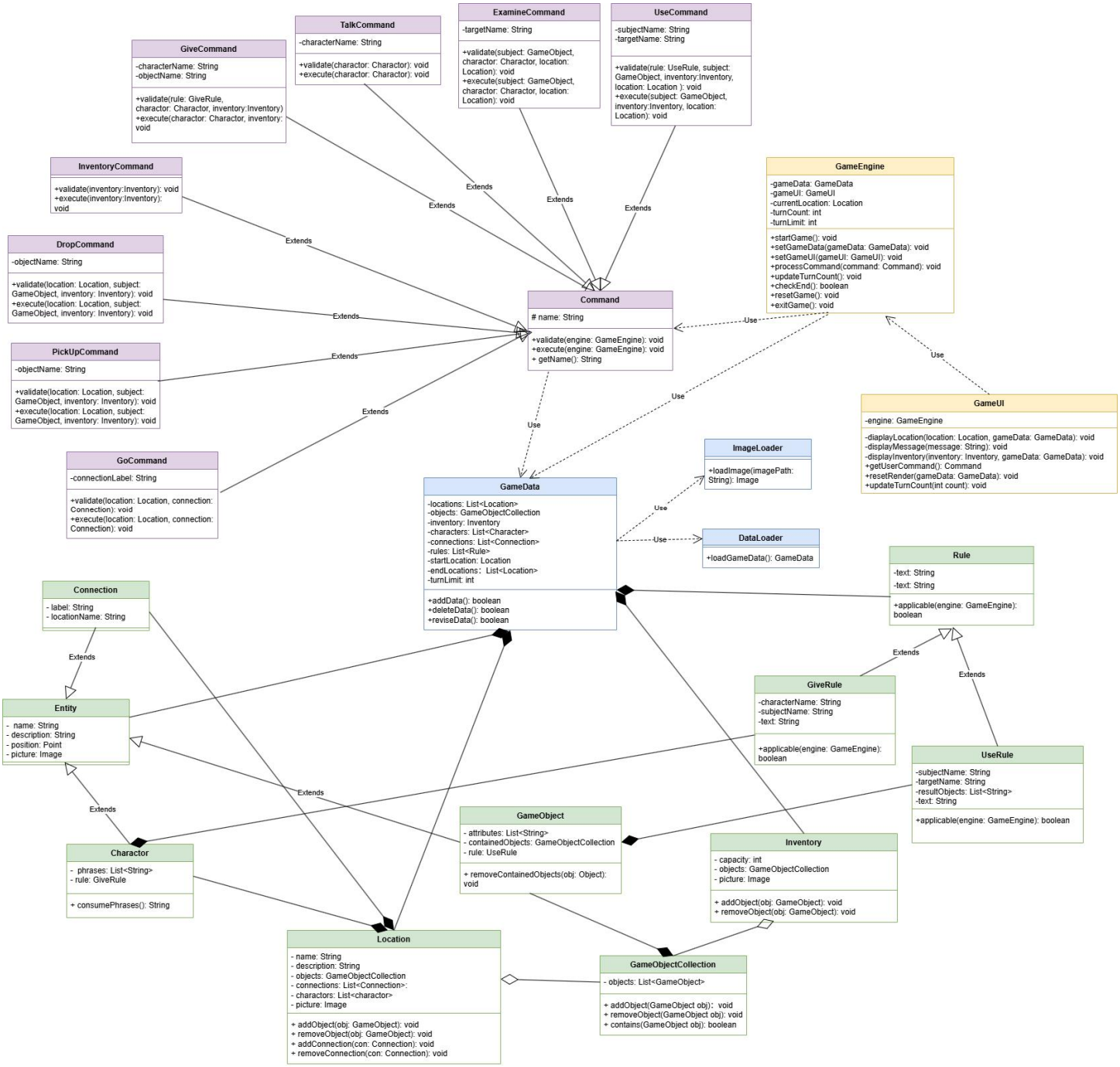
- **GameEngine**: Central controller coordinating game data, UI, command execution, and turn updates.
- **GameUI**: Interface bridge that displays the current location, inventory, turn count, command bar and player messages.
- **ImageLoader** and **DataLoader**: Utility classes for loading images and parsing external game data files.
- **GameData**: Container for the entire game world, including locations, entities, rules and player inventory.
- **Location**: Represents a game area containing objects, characters and connecting paths.
- **Connection**: Directional link between locations; a specialized entity enabling player movement.
- **GameObject**: All interactive items, supporting contained objects and attributes.
- **Entity(abstract)**: Connections, characters and gameobjects can be considered as a kind of entity in this game.
- **Rule (abstract)** and subclasses **UseRule** and **GiveRule**: Implement game logic conditions and effects.
- **Command (abstract)**: Defines player actions with subclasses like **GoCommand**, **PickupCommand**, **UseCommand**, each implementing validation execution.
- **Inventory**: Manages the collection of objects held by the player, supporting add/remove operations essential for gameplay consistency.

Relationships

- **Composition:**
  - **GameData** is composed of locations, rules, entities and inventory.
  - Each **Location** contains collections of charactors and connections.
  - **Character** and **GiveRule** are bundled.
- **Aggregation:**
  - **Location** and **Inventory** aggregates **GameObjectCollection**.
- **Inheritance:**
  - **GameObject**, **Connection**, and **Character** inherit from **Entity**.
  - **UseRule** and **GiveRule** extend abstract **Rule**.
  - All specific commands derive from **Command**.
- **Dependency:**
  - **GameEngine** interacts with **GameData**, **Command** and **GameUI**.
  - **ImageLoader** and **DataLoader** was called by **GameData**.

Design Highlights

- Adoption of the **Command Pattern** enables extensible and testable player actions.
- A **Data-Driven Architecture** separates configuration (loaded externally) from runtime logic.
- **Rule abstraction** simplifies implementation of conditional game mechanics.
- Encapsulation of game state within **GameEngine** improves maintainability and reduces side effects.
- Modular subsystems facilitate independent testing and future expansions.


GiveCommand
-characterName: String
-objectName: String
+validate(rule: GiveRule, charactor: Charactor, inventory:Inventory)
+execute(charactor: Charactor, inventory: void
TalkCommand
-characterName: String
+validate(charactor: Charactor): void
+execute(charactor: Charactor): void
ExamineCommand
-targetName: String
+validate(subject: GameObject, charactor: Charactor, location: Location): void
+execute(subject: GameObject, charactor: Charactor, location: Location): void
UseCommand
-subjectName: String
-targetName: String
+validate(rule: UseRule, subject: GameObject, inventory:Inventory, location: Location): void
+execute(subject: GameObject, inventory:Inventory, location: Location): void
InventoryCommand
+validate(inventory:Inventory): void
+execute(inventory:Inventory): void
DropCommand
-objectName: String
+validate(location: Location, subject: GameObject, inventory: Inventory): void
+execute(location: Location, subject: GameObject, inventory: Inventory): void
PickUpCommand
-objectName: String
+validate(location: Location, subject: GameObject, inventory: Inventory): void
+execute(location: Location, subject: GameObject, inventory: Inventory): void
GoCommand
-connectionLabel: String
+validate(location: Location, connection: Connection): void
+execute(location: Location, connection: Connection): void
Extends
Command
name: String
+validate(engine: GameEngine): void
+execute(engine: GameEngine): void
+ getName(): String
Use
GameEngine
-gameData: GameData
-gameUI: GameUI
-currentLocation: Location
-turnCount: int
-turnLimit: int
+startGame(): void
+setGameData(gameData: GameData): void
+setGameUI(gameUI: GameUI): void
+processCommand(command: Command): void
+updateTurnCount(): void
+checkEnd(): boolean
+resetGame(): void
+exitGame(): void
GameUI
-engine: GameEngine
-diaplayLocation(location: Location, gameData: GameData): void
-displayMessage(message: String): void
-displayInventory(inventory: Inventory, gameData: GameData): void
+getUserCommand(): Command
+resetRender(gameData: GameData): void
+updateTurnCount(int count): void
ImageLoader
+loadImage(imagePath: String): Image
DataLoader
+loadGameData(): GameData
GameData
-locations: List<Location>
-objects: GameObjectCollection
-inventory: Inventory
-characters: List<Character>
-connections: List<Connection>
-rules: List<Rule>
-startLocation: Location
-endLocations：List<Location>
-turnLimit: int
+addData(): boolean
+deleteData(): boolean
+reviseData(): boolean
Rule
-text: String
-text: String
+applicable(engine: GameEngine): boolean
Connection
- label: String
- locationName: String
Entity
- name: String
- description: String
- position: Point
- picture: Image
GiveRule
-characterName: String
-subjectName: String
-text: String
+applicable(engine: GameEngine): boolean
UseRule
-subjectName: String
-targetName: String
-resultObjects: List<String>
-text: String
+applicable(engine: GameEngine): boolean
GameObject
- attributes: List<String>
- containedObjects: GameObjectCollection
- rule: UseRule
+ removeContainedObjects(obj: Object): void
Charactor
- phrases: List<String>
- rule: GiveRule
+ consumePhrases(): String
Inventory
- capacity: int
- objects: GameObjectCollection
- picture: Image
+ addObject(obj: GameObject): void
+ removeObject(obj: GameObject): void
Location
- name: String
- description: String
- objects: GameObjectCollection
- connections: List<Connection>:
- charactors: List<charactor>
- picture: Image
+ addObject(obj: GameObject): void
+ removeObject(obj: GameObject): void
+ addConnection(con: Connection): void
+ removeConnection(con: Connection): void
GameObjectCollection
- objects: List<GameObject>
+ addObject(GameObject obj)： void
+ removeObject(GameObject obj): void
+ contains(GameObject obj): boolean

Comments

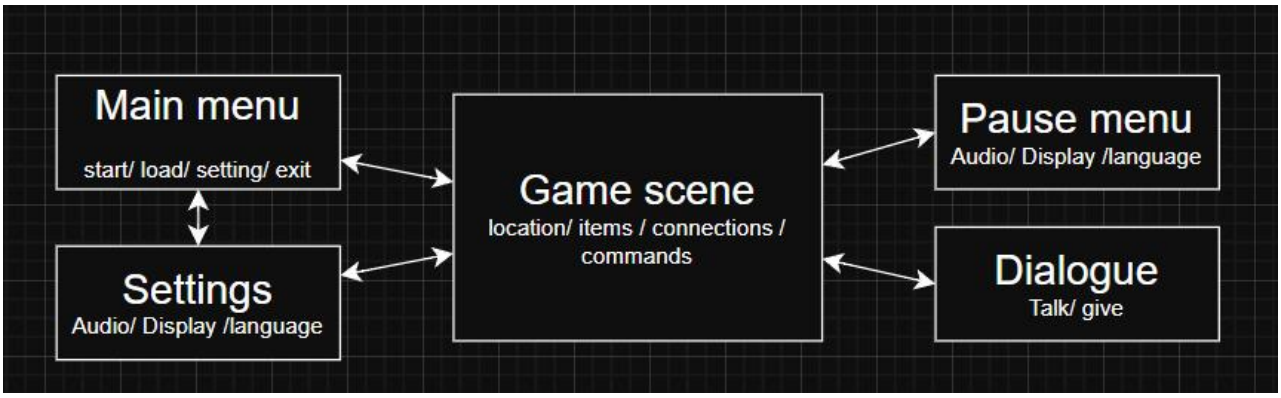
User Interface Design Documentation

1. Overview

This UI defines a visual, point-and-click layer over classic text-adventure mechanics. Players act through clear on-screen elements rather than typing, while puzzle depth is preserved. The interface favors clarity, immediate feedback, and consistent interactions. All actions should be discoverable, all results visible, and no operation should fail silently.

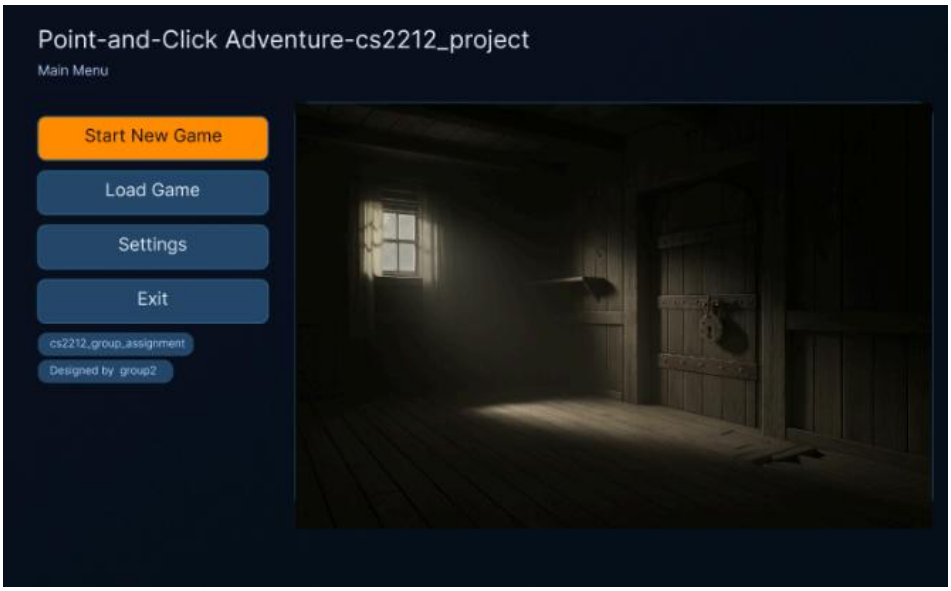
2. Navigation Flow

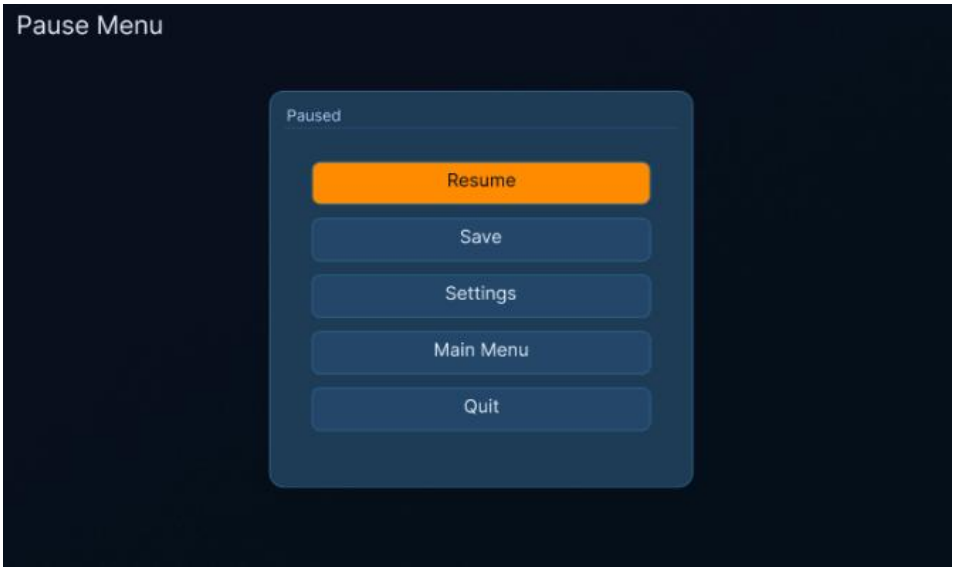
On launch, the application loads game data from JSON or XML and then shows the Main Menu. New Game leads to an initial starting message and then the Game Main Interface; Load Game (if present) resumes directly into the Game Main Interface; Settings returns to the menu after changes; Exit closes the application. During play, the user can open Settings or Pause if provided, then return to the Game Main Interface without losing context.



3. Main Menu

The application opens on the Main Menu, which sets the visual tone and provides primary navigation. It displays the title or logo and offers New Game, optional Load Game, Settings, and Exit. From here the user can start a new session, return to an existing one if implemented, adjust configuration, or quit.





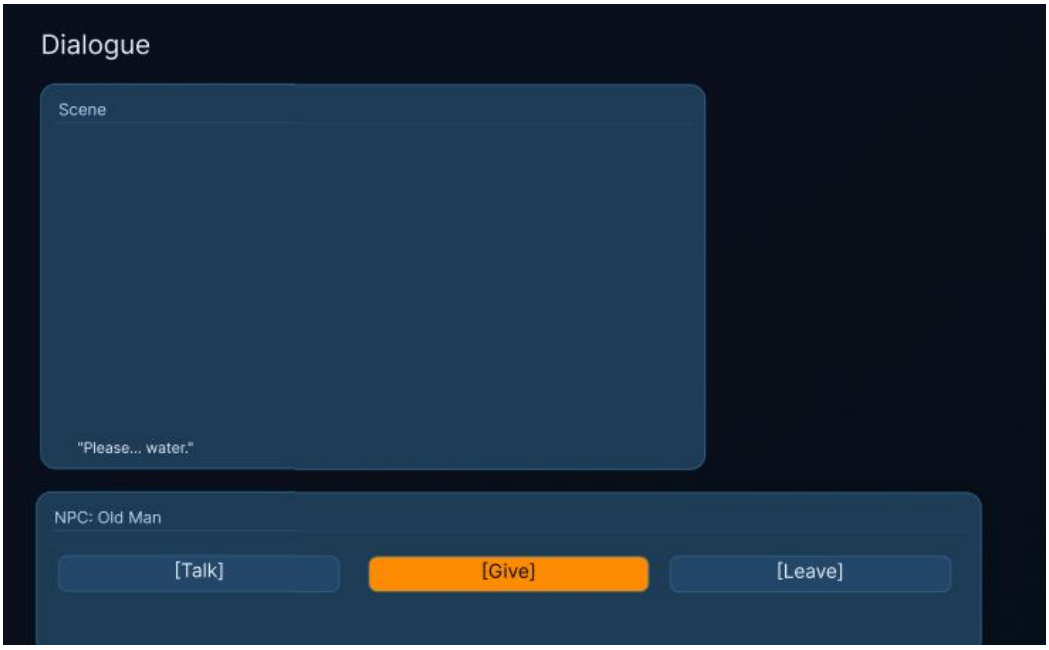
4. Game Scene

Design Rationale

Area	Function	Design Principle
Upper Game Scene	Displays the current room, characters, and objects.	Maintains player immersion by simulating a theatrical stage perspective. The scene provides strong spatial cues (doors, stairs, items, etc.).
Middle Command Verbs Area	Contains a set of clickable verb commands such as "Pick," "Use," "Talk to," etc.	Replaces text-based command input with a graphical interface, making interactions more intuitive.
Lower Inventory Area	Displays the items currently held by the character.	Encourages players to think in terms of "item + action" combinations, forming a visual interactive language.

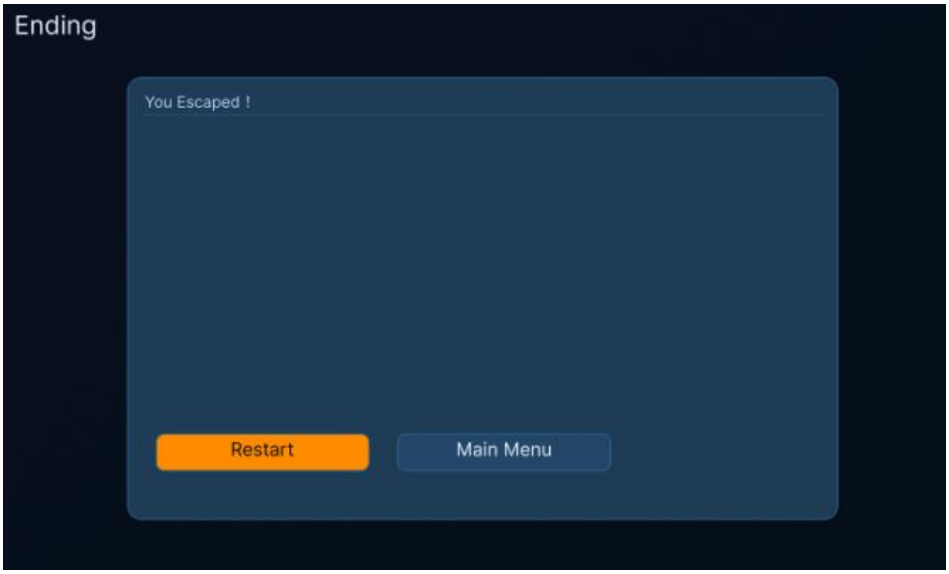
5. Dialogue Interface

When talking to a character, a semi-transparent overlay appears that preserves the sense of place. It shows the character name, an optional portrait, and the current line of dialogue. A Continue or Next control advances the sequence, while Close exits the conversation. When a character has no more to say, the interface explicitly indicates that the dialogue is exhausted.



6. End Screen

Reaching an ending location or hitting a turn limit transitions to an End Screen. It summarises the outcome with a victory or defeat message and shows the final turn count. From here the player may Play Again to restart, return to the Main Menu, or Exit the application.



Comments

4 Summary

This project delivers a data-driven, Java desktop point-and-click adventure engine whose UI replaces free-form text parsing with clear, consistent on-screen actions. At runtime the engine loads JSON or XML to define locations, objects, characters, connections, and interaction rules; content lives outside the codebase while the UI presents a three-area Game Main Interface (scene display, fixed command panel, feedback/status), alongside a Main Menu, Inventory panel, Dialogue overlay, Settings, and an End Screen. Core commands—Go, Pick Up, Drop, Examine, Use, Talk, Give, and optionally Inventory—follow a “command-first, then target” pattern, and multi-object actions use lightweight parameter dialogs. Feedback is immediate and visible: the log prints results, the scene and inventory update atomically, and a prominent turn counter advances with optional limits; navigation is straightforward from launch to New Game, with Load and Settings always accessible and a clean return path during play. Non-functional requirements emphasize Java (Swing/JavaFX), robust JSON/XML parsing, Javadoc documentation, responsive performance, and team coding standards to keep the codebase reliable and comprehensible.

The underlying architecture, as captured in the class diagrams, separates concerns into loosely coupled yet cohesive subsystems. `GameEngine` orchestrates flow, advances turns, and mediates between data, commands, and UI; `GameData` composes the world and relies on `DataLoader` and `ImageLoader` for external assets. The object and rule model centers on an `Entity` abstraction specialized by `GameObject`, `Connection`, and `Character`, while an abstract `Rule` is realized by `UseRule` and `GiveRule` to encapsulate conditions and effects. Player actions use the Command Pattern: an abstract `Command` defines validation and execution and is extended by concrete commands such as `GoCommand`, `PickupCommand`, `DropCommand`, `ExamineCommand`, `UseCommand`, `TalkCommand`, `GiveCommand`, and `InventoryCommand`. Key relationships include composition of world elements within `GameData`, aggregation of `GameObjectCollection` by `Location` and `Inventory`, inheritance from `Entity` for all interactive elements, and controlled dependencies from `GameEngine` to `GameData`, `Command`, and `GameUI`. Together, these choices preserve puzzle-solving depth while improving discoverability and usability, and they cleanly separate content from engine logic to enable extensibility, testability, and long-term maintainability.

Comments